

COMPETITION MATHEMATICS • GRADES 8 AND BELOW

AMC 8 Comprehensive Cheat Sheet

Core knowledge, high-yield methods, and Q20–25 problem-solving tools — rebuilt for complete AMC 8 coverage.

Official format

- **25** multiple-choice questions
- **40 minutes**
- **1 point** per correct answer
- **0 penalty** for wrong/blank

Permitted mindset

- Calculators are not permitted
- Ruler + scratch paper are useful
- Figures may **not** be to scale
- Estimate before heavy algebra

Time target

- Q1–10: about **10 min**
- Q11–20: about **18 min**
- Q21–25 + review: **12 min**
- Never leave a bubble blank

0 Test-day strategy CORE**High-ROI moves**

- **Skip fast:** if stuck for 60–90 seconds, mark it and move.
- **Eliminate:** use size, parity, units, signs, or bounds.
- **Estimate:** round aggressively when answer choices are separated.
- **Backsolve:** plug answer choices into the condition.
- **Try small cases:** use $n = 1, 2, 3$ to expose a pattern.

Before committing

- Draw and label the picture yourself.
- Check what is actually asked: **area vs perimeter**, value vs count, probability vs number of outcomes.
- Ask whether there is a faster route: symmetry, complement, factorization, or working backwards.
- For geometry, never trust apparent lengths/angles from the sketch.

1 Arithmetic & Number Sense CORE

Fractions • decimals • percents

$$\frac{1}{2} = 0.5, \quad \frac{1}{4} = 0.25, \quad \frac{3}{4} = 0.75$$
$$\frac{1}{5} = 0.2, \quad \frac{1}{8} = 0.125, \quad \frac{1}{10} = 0.1$$

Percent change:

$$\text{new} = \text{old} \left(1 \pm \frac{p}{100} \right)$$

Compare fractions: for positive denominators, compare ***ad*** and ***bc*** in $\frac{a}{b}$ and $\frac{c}{d}$.

Reverse percentage: if final = **120%** of original, divide by **1.2**, not subtract **20%**.

Fast arithmetic

- **Distribute:** $49 \cdot 18 = (50 - 1)18$.
- **Difference of squares:**
 $a^2 - b^2 = (a - b)(a + b)$.
- **Order of operations:** brackets → exponents → multiply/divide → add/subtract.
- **Absolute value:** $|x|$ is distance from **0**, so never negative.
- **Arithmetic list mean:** for equally spaced terms, mean = $\frac{\text{first} + \text{last}}{2}$.

Know these instantly

Squares:

1, 4, 9, 16, 25, 36, 49, 64, 81, 100

121, 144, 169, 196, 225, 256, 289, 324, 361, 400

Cubes:

1, 8, 27, 64, 125

216, 343, 512, 729, 1000

Units: write them. Many AMC 8 choices are designed to trap a correct number with the wrong unit.

2 Ratios, Proportions & Rates CORE

Equivalent ratios

$$\frac{a}{b} = \frac{c}{d} \iff ad = bc$$

- Ratio $a : b$ has $a + b$ total parts.
- If $a : b = 3 : 5$ and total is **64**, one part is $64/8 = 8$.
- Part-to-whole:** if red:blue = $2 : 3$, red is $\frac{2}{5}$ of the total.
- When a quantity is added/removed, write the **new** ratio explicitly.

Rates & scaling

$$\text{unit rate} = \frac{\text{quantity}}{1 \text{ unit}}, \quad d = rt$$

- Unit price, speed, density, pages/minute are all rates.
- Scale factor k : lengths multiply by k , areas by k^2 , volumes by k^3 .
- Direct proportion:** $y = kx$.
- Convert units **before** comparing rates.

3 Algebra Essentials CORE

Translate words → equations

- Consecutive integers: $n, n + 1, n + 2, \dots$
- Even integers: $2k$; odd integers: $2k + 1$
- Two-digit number: $10a + b$
- Reverse digits: $10b + a$
- " x more than y " means $y + x$.
- " x less than y " means $y - x$.

Linear equations

Expand carefully, combine like terms, then isolate the variable. Check by substitution if the algebra was messy.

Systems

For two unknowns, use substitution or elimination. Word problems often become two equations in two variables.

Exponents & identities

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn}$$

$$(ab)^n = a^n b^n, \quad a^0 = 1 \quad (a \neq 0)$$

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

$$a^2 - b^2 = (a - b)(a + b)$$

Watch negatives: $(-3)^2 = 9$, but $-3^2 = -9$.

Distance, work & mixtures

Distance: $d = rt$.

Relative speed: toward each other → add speeds; same direction → subtract.

Work: if one job takes t units of time, rate = $\frac{1}{t}$.

$$\frac{1}{T} = \frac{1}{t_1} + \frac{1}{t_2} + \dots$$

Mixture / weighted average:

$$\text{average} = \frac{\sum(\text{amount})(\text{value})}{\sum \text{amount}}$$

Sequences & patterns HIGH

Arithmetic:

$$a_n = a_1 + (n - 1)d, \quad S_n = \frac{n}{2}(a_1 + a_n)$$

Geometric:

$$a_n = a_1 r^{n-1}$$

- For repeated operations, compute the first few terms.
- Look for a cycle, fixed point, parity pattern, or invariant.
- For periodic sequences, reduce the index modulo the cycle length.

4 Data, Graphs & Tables CORE

Center & spread

$$\text{mean} = \frac{\text{sum}}{\text{count}}$$

- **Median:** middle after sorting.
- **Mode:** most frequent value.
- **Range:** maximum — minimum.
- If one data value changes by k , the total sum changes by k .
- If mean of n values is m , total sum is nm .

Weighted mean

$$\bar{x} = \frac{n_1x_1 + n_2x_2 + \cdots}{n_1 + n_2 + \cdots}$$

Do **not** average group averages unless the groups have equal size.

Frequency table shortcut

If value x occurs f times, contribution to the sum is fx .

If the graph is crowded, reconstruct the underlying table before calculating.

Know the graph types

- **Bar chart:** separate categories.
- **Histogram:** numerical intervals; bars normally touch.
- **Line graph:** change over an ordered variable, often time.
- **Scatter plot:** relationship/correlation between two variables.
- **Pie chart:** sector angle = $\text{fraction} \times 360^\circ$.

$$\text{sector fraction} = \frac{\theta}{360^\circ}$$

Graph-reading traps

- Read the **axis scale**; it may not increase by 1.
- Check whether axes start at zero.
- Distinguish frequency from cumulative frequency.
- Use slope as a **rate of change**.
- For a trend, do not assume every point lies on a perfect line.

Histogram warning: bars represent intervals, not individual categories.

5 Coordinate Geometry & Beginning Functions HIGH

Coordinate essentials

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- Horizontal line: slope **0**.
- Vertical line: slope undefined.
- Distance horizontally/vertically is just coordinate difference.

Linear & quadratic ideas

$$y = mx + b$$

- m = rate of change / slope.
- b = y -intercept.
- A constant first difference suggests a linear rule.
- A constant second difference suggests a quadratic pattern.
- For tables, inspect differences before guessing the formula.

Fast function-thinking

Input/output:

A function assigns one output to each allowed input.

Intersection:

A point where two graphs meet satisfies both rules.

Bounds:

If choices are far apart, estimate from the graph before solving exactly.

6 Geometry Formula Bank I CORE

Angles

- Straight line: 180°
- Around a point: 360°
- Vertical angles are equal.
- Complementary angles sum to 90° .
- Supplementary angles sum to 180° .
- Parallel lines: corresponding and alternate interior angles are equal.

Triangles

- Interior sum: 180°
- Isosceles triangle: base angles equal.
- Triangle inequality: each side is less than the sum of the other two.

Perimeter & area

Shape	Formula
Rectangle	$P = 2(L + W), \quad A = LW$
Triangle	$A = \frac{1}{2}bh$
Parallelogram	$A = bh$
Trapezoid	$A = \frac{1}{2}(b_1 + b_2)h$
Circle	$C = 2\pi r, \quad A = \pi r^2$

Composite figures: split into simple pieces or subtract missing pieces.

Same altitude: triangle areas are proportional to their bases.

Right triangles

$$a^2 + b^2 = c^2$$

Triples:

3–4–5, 5–12–13, 8–15–17, 7–24–25,
plus multiples.

If the hypotenuse and one leg are known, square first;
do not approximate too early.

Polygons

$$\begin{aligned}\text{interior angle sum} &= (n - 2)180^\circ \\ \text{exterior angle sum} &= 360^\circ\end{aligned}$$

For a regular n -gon:

$$\begin{aligned}\text{exterior angle} &= \frac{360^\circ}{n} \\ \text{interior angle} &= 180^\circ - \frac{360^\circ}{n}\end{aligned}$$

Central angle of a regular n -gon is also $\frac{360^\circ}{n}$.

Circles: high-yield facts

$$\begin{aligned}L_{\text{arc}} &= \frac{\theta}{360^\circ}(2\pi r) \\ A_{\text{sector}} &= \frac{\theta}{360^\circ}(\pi r^2) \\ A_{\text{annulus}} &= \pi(R^2 - r^2)\end{aligned}$$

- Radius is perpendicular to tangent at the point of tangency.
- Equal tangents from the same external point have equal lengths.
- Inscribed angle = $\frac{1}{2}$ central angle subtending the **same arc**.

Special right triangles HIGH

$45^\circ - 45^\circ - 90^\circ$:

$$x : x : x\sqrt{2}$$

$30^\circ - 60^\circ - 90^\circ$:

$$x : x\sqrt{3} : 2x$$

7 Geometry Formula Bank II CORE

Similarity & scale

Similar triangles have equal corresponding angles and proportional corresponding sides.

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = k$$

$$\frac{A_1}{A_2} = k^2, \quad \frac{V_1}{V_2} = k^3$$

- A line parallel to one side of a triangle creates similar triangles.
- Keep correspondence consistent when writing proportions.

Transformations & symmetry

- **Translation:** slide; lengths/angles unchanged.
- **Reflection:** mirror across a line.
- **Rotation:** turn about a center.
- **Dilation:** scale by factor k .

Quarter-turn about origin:

$$(x, y) \xrightarrow{90^\circ \text{ CCW}} (-y, x)$$

For visual problems, track one distinctive vertex/edge rather than the whole figure.

Area-ratio weapons

- **Same height:** areas $\propto$ bases.
- **Same base:** areas $\propto$ heights.
- A median divides a triangle into **two equal-area** triangles.
- The three medians divide a triangle into **6 equal-area** small triangles.

$$\frac{[\triangle_1]}{[\triangle_2]} = \frac{\text{base}_1}{\text{base}_2} \quad (\text{same height})$$

3D geometry & spatial visualization

Solid	Key fact
Cube	6 faces, 12 edges, 8 vertices
Rectangular prism	$V = lwh$
Cube	$V = s^3, \quad SA = 6s^2$
Rectangular prism	$SA = 2(lw + lh + wh)$

- For painted-cube problems, classify corner/edge/face/interior cubes.
- For nets, mentally fold one face at a time.
- Hidden blocks still contribute to volume even when not visible.

Common geometry tactics

Add a line

Draw a diagonal, altitude, radius, or extension.

Break the shape

Split into rectangles/triangles or subtract gaps.

Use symmetry

Analyze half or one repeated sector, then multiply.

Coordinates

Place a difficult figure on a grid if that simplifies lengths.

Area instead of length

Ratios are often easier through area.

Do not trust scale

Use only stated or proven relationships.

8 Number Theory & Remainders CORE

Primes & factorization

- A prime has exactly two positive divisors: **1** and itself.
- 2** is the only even prime.
- 1** is **not** prime.
- Prime factorization is the fastest route to GCD, LCM, divisor count, and square/cube tests.

$$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$$

Perfect square: every prime exponent is even.

Perfect cube: every prime exponent is a multiple of 3.

GCD & LCM

Using prime exponents:

- gcd:** take the **minimum** exponent.
- lcm:** take the **maximum** exponent.

$$\gcd(a, b) \cdot \text{lcm}(a, b) = ab$$

For two positive integers only. Prime factorization is usually safer than memorizing algorithms at AMC 8 level.

Remainders & modular thinking

" n leaves remainder r when divided by m " means

$$n = mq + r, \quad 0 \leq r < m$$

Example: remainder **2** mod **5** $\rightarrow n = 5k + 2$.

- Days of week $\rightarrow$ work mod **7**.
- Last digit $\rightarrow$ work mod **10**.
- Parity $\rightarrow$ work mod **2**.

Divisibility tests

Divisor	Test
2	last digit even
3	digit sum divisible by 3
4	last 2 digits divisible by 4
5	ends in 0 or 5
6	divisible by both 2 and 3
8	last 3 digits divisible by 8
9	digit sum divisible by 9
10	ends in 0
11	alternating digit sum is a multiple of 11

Counting divisors

$$n = p_1^{e_1} p_2^{e_2} \cdots$$

$$d(n) = (e_1 + 1)(e_2 + 1) \cdots$$

Useful consequence: a positive integer has an odd number of positive divisors exactly when it is a perfect square.

Counting multiples: multiples of m from **1** through N :

$$\left\lfloor \frac{N}{m} \right\rfloor$$

Cycles & last digits

Powers often repeat in cycles.

Last digits of powers of **2: 2, 4, 8, 6** (period 4).

Powers of **3: 3, 9, 7, 1** (period 4).

Powers of **7: 7, 9, 3, 1** (period 4).

Powers of **9: 9, 1** (period 2).

Method: find the cycle length, reduce the exponent modulo that length, then read the corresponding position.

9 Counting & Probability CORE

Core counting tools

- **Product rule:** do A then B → multiply choices.
- **Sum rule:** either disjoint case A or B → add.
- **Casework:** split by a clean feature and add.
- **Complement:** total — unwanted.
- **Ordered list:** excellent when the space is small.

AMC habit: before using a formula, ask whether simple casework would be shorter.

Factorials, permutations, combinations

$$n! = n(n-1) \cdots 2 \cdot 1$$

$${}_nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Order matters: permutation.

Order does not matter: combination.

Probability

When elementary outcomes are **equally likely**:

$$P(E) = \frac{\# \text{ favorable outcomes}}{\# \text{ total outcomes}}$$

Complement:

$$P(\text{at least one}) = 1 - P(\text{none})$$

Without replacement: probabilities change after each draw.

Independent events: multiply probabilities when both must occur.

Grid paths

From $(0, 0)$ to (a, b) using only right/up moves, every path has $a + b$ moves: a R's and b U's.

$$\# \text{paths} = \binom{a+b}{a} = \binom{a+b}{b}$$

If a path must pass through a point, count paths to the point and multiply by paths from the point to the destination.

Repeated objects & overcounting

For a string of n objects with repeated groups of sizes $a, b, c, \dots$:

$$\frac{n!}{a!b!c!\cdots}$$

Only divide by repeat-factorials when the overcount really comes from swapping identical objects.

Symmetry correction

If each final object is counted exactly k times, divide by k .

Inclusion–exclusion

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Use for “divisible by 2 or 3,” “likes A or B,” etc.

Pigeonhole Principle HIGH

If N objects go into k boxes, some box contains at least

$$\left\lceil \frac{N}{k} \right\rceil$$

Common counting questions

“At least one”

Try complement.

“No two adjacent”

Arrange gaps or use casework.

“How many numbers?”

Count digit choices carefully; first digit cannot be 0.

10 High-Frequency Word-Problem Templates CORE

Ages

- If current age is x , then t years ago: $x - t$; in t years: $x + t$.
- Age **differences stay constant**.
- Ratios of ages do not stay constant.

Coins / money

Use two equations:

number equation, **value equation**

Digits

Two-digit number = $10a + b$; reversed = $10b + a$.
Digit constraints: $0 \leq a, b \leq 9$, and a leading digit is nonzero.

Calendars, clocks & cycles

- Weekdays repeat every **7** days.
- Clock face: $360^\circ / 12 = 30^\circ$ between hour marks.
- Minute hand: **6°** per minute.
- Hour hand: **0.5°** per minute.

For "what day is it after N days?" reduce N modulo 7.

Speed / travel

$$d = rt$$

- Same distance, different speeds $\rightarrow$ times are inversely related.
- Average speed is $\frac{\text{total distance}}{\text{total time}}$, not usually the average of speeds.
- Toward each other $\rightarrow$ relative speed adds.

Work / filling

Add **rates**, not completion times.

Mixtures

Track amount of the ingredient, not just the percentage.

Units & conversions

- Area units square: $1 \text{ m}^2 = 10,000 \text{ cm}^2$.
- Volume units cube: $1 \text{ m}^3 = 1,000,000 \text{ cm}^3$.
- Time: **60** sec = **1** min; **60** min = **1** hr.
- Always convert to compatible units before using $d = rt$ or a density/rate.

Classic trap: multiplying a length conversion by itself or itself three times for area/volume.

11 Logic & Process Problems HIGH

Tables & constraints

- For matching/scheduling problems, make a compact table.
- Mark impossible pairings first.
- Use the strongest constraint before trying cases.
- If integers must be positive/distinct, write that constraint visibly.

Games & repeated moves

- Track parity, total sum, color counts, or remainders.
- Work backwards if each move can be reversed.
- Look for a quantity that never changes.
- Try a smaller version before attacking the full process.

Geometry in disguise

A fence problem may ask perimeter. A tiled floor asks area. A painted box may ask exposed surface area. A shortest route may become a grid-path count. Translate the story into a mathematical object before calculating.

ADVANCED AMC 8 CONCEPTS

Q20–25 Power Tools

These are not “extra formulas to memorize.” Use them when a hard problem has structure that the basic methods do not expose quickly.

1. Area-ratio reasoning

If two triangles share an altitude:

$$\frac{[\triangle_1]}{[\triangle_2]} = \frac{\text{base}_1}{\text{base}_2}$$

If they share a base:

$$\frac{[\triangle_1]}{[\triangle_2]} = \frac{h_1}{h_2}$$

This is often faster than solving several lengths.

2. Symmetry & equal sharing

- A median gives two equal-area triangles.
- Rotational/reflection symmetry can force equal counts or areas.
- If identical sectors repeat, solve one sector and multiply.

Before using algebra, ask: “Is this figure secretly made of equal pieces?”

3. Pigeonhole Principle

More objects than boxes forces a repeat.

Typical AMC uses:

- same remainder
- same month/day category
- two numbers with related difference
- at least two sharing a property

$$\text{some box has at least } \left\lceil \frac{N}{k} \right\rceil$$

4. Invariants

An invariant is a quantity that **does not change** under the allowed move.

- parity
- sum modulo 2, 3, ...
- number of black/white objects modulo 2
- total value

If start and target have different invariant values, the target is impossible.

5. Extremal principle

Choose the largest, smallest, earliest, latest, leftmost, etc., then exploit what must be true about it.

Very useful when “all objects” are hard to analyze simultaneously.

6. Work backwards

If operations can be undone, start from the final state.

$$+a \leftrightarrow -a, \quad \times b \leftrightarrow \div b$$

Often turns a multi-step process into one straight chain.

7. Double counting

Count the same set of objects in two different ways.

Classic example:

$$\text{handshakes} = \frac{\text{sum of degrees}}{2}$$

Use when a total can be viewed “by people” and “by pairs,” or “by rows” and “by columns.”

8. Bounds before exact work

- Find a minimum/maximum.
- Estimate the order of magnitude.
- Compare against widely spaced options.

If only one choice lies inside the proven range, stop. Exact computation is unnecessary.

12 Selective Bonus Tools ONLY WHEN USEFUL

Angle Bisector Theorem

If AD bisects $\angle A$ in $\triangle ABC$ and $D \in BC$:

$$\frac{BD}{DC} = \frac{AB}{AC}$$

Worth knowing, but lower priority than similarity, area ratios, sectors, and spatial reasoning.

Intersecting chords

If chords AB and CD intersect at P inside a circle:

$$PA \cdot PB = PC \cdot PD$$

Use only when a circle problem naturally creates a product of segment lengths.

Algebra structure spotting

- Factor repeated chunks: $ab + ac = a(b + c)$.
- Look for $a^2 - b^2$ before expanding.
- If the same complicated expression repeats, set it equal to a new variable.
- Use symmetry: expressions in x and $1/x$ may simplify after squaring a sum.

Stars & Bars RARE

For nonnegative integer solutions to

$$x_1 + x_2 + \cdots + x_k = N$$

$$\# = \binom{N + k - 1}{k - 1}$$

Do not reach for this unless the problem is genuinely distributing identical objects into labeled groups.

✓ Final 20-Second Checklist

- ☐ Did I answer the quantity actually asked for?
- ☐ Are my units correct?
- ☐ Is the sign/value physically or geometrically reasonable?
- ☐ Did I accidentally use diameter where I needed radius?
- ☐ Did I square/cube the scale factor for area/volume?
- ☐ For probability, were the outcomes equally likely?

- ☐ For median, did I sort first?
- ☐ For average speed, did I use total distance / total time?
- ☐ For a diagram, did I assume anything not given?
- ☐ For ratios, did I distinguish part-to-part from part-to-whole?
- ☐ Does exactly one answer choice match?
- ☐ If I am out of time, have I guessed every remaining question?

★ What to Memorize Cold

Absolutely automatic	Recognize quickly	Use strategically
Fraction-percent conversions $d = rt$ triangle/circle/rectangle area Pythagorean theorem slope divisibility tests	common Pythagorean triples perfect squares/cubes arithmetic sequences last-digit cycles similarity scale k, k^2, k^3	complement counting pigeonhole area ratios invariants work backwards symmetry/bounds

Best AMC 8 habit: do not ask "Which formula fits?" first. Ask "What structure is hiding here?" Then choose the shortest valid method.

Coverage note

This reference is organized around the current MAA AMC 8 description: counting and probability, estimation, proportional reasoning, elementary geometry including the Pythagorean Theorem, spatial visualization, and interpreting graphs and tables; later questions may also touch beginning linear/quadratic functions and coordinate geometry. Test-format and permitted-material notes reflect current MAA AMC rules.